



EcoKonek

A Web-Based Solution for Responsible E-waste Disposal and Bridging the Digital Gap in Local Communities in Mexico Pampanga

ELECTRONICS

Abstract

EcoKonek PH is a web-based platform developed to tackle the dual challenges of rapid e-waste growth and the digital divide in the Philippines. At its core, the system utilizes an AI-powered chatbot to educate the public and streamline the donation of functional electronics to underserved communities. By integrating features like barangay-level mapping and a real-time environmental impact dashboard, the platform provides a centralized hub for households, local government units (LCUs), and NGOs to collaborate on sustainable waste management. The development of the platform followed Agile and Scrum methodologies, ensuring the final product was highly responsive to user needs. A quantitative study involving 50 residents in Mexico, Pampanga, validated the system's effectiveness, yielding a 98.1% satisfaction rating for usability and accessibility. This research highlights a strong communal willingness to engage in responsible disposal practices when given the right tools. Ultimately, EcoKonek PH serves as a scalable model that aligns with global sustainability goals, demonstrating how smart technology can successfully bridge the gap between environmental advocacy and social equity.

Keywords: E-waste management, Digital inclusion, Ekokonek, Green computing, Gamified, E-waste recycling, Sustainable Technology

Objectives



To design and develop EcoKonek PH, a web-based platform that supports responsible e-waste disposal through features such as barangay-level mapping, verified drop-off points, and a donation and redistribution system.



To promote digital inclusion by enabling the redistribution of functional electronic devices from donating households to underserved users such as students, seniors, and persons with disabilities.



To validate community adoption readiness by measuring the system's ease of use, perceived usefulness, and overall user trust.



Results



HIGH PERCEIVED USEFULNESS



SUCCESSFUL SYSTEM EVALUATION



VALID COMMUNITY ACCEPTANCE



EFFECTIVE ECOMONEK PH FEATURES



EFFECTIVE DIGITAL INCLUSION MODEL

Conclusion

EcoKonek PH successfully bridges the gap between environmental sustainability and digital equity. By integrating AI-driven tools with a user-centered design, the platform has established a secure and trusted system for e-waste disposal and technology donation. The pilot implementation resulted in a measurable improvement in proper waste handling, increased digital inclusion, and a stronger culture of community stewardship. EcoKonek PH stands as a proven, scalable model for fostering a cleaner, more digitally connected future.

Methods

Research Design:



Uses a descriptive quantitative research design to measure community e-waste practices, digital access, and user perceptions of EcoKonek PH through a structured online survey.

System Development:



Follows an Agile development framework with Scrum, implementing EcoKonek PH in iterative sprints to build the AI chatbot, barangay mapping, donation system, and impact dashboard based on continuous user and stakeholder feedback.

Participants & Setting:



Involves 50 household residents of Mexico, Pampanga as survey respondents, along with key stakeholders such as barangay officials and environmental officers who inform system requirements and community context.

Research Instruments:



Utilizes a TAM-aligned usability questionnaire administered via Google Forms to assess perceived usefulness, ease of use, attitude/trust, and behavioral intention toward EcoKonek PH, supplemented by guide questions for stakeholders on local e-waste issues.

Data Analysis:



Applies descriptive statistics (means and standard deviations) to summarize user responses for each item and interpret overall acceptability, integrating these results with stakeholder feedback to evaluate EcoKonek PH's effectiveness for responsible e-waste management and digital inclusion.

Discussion

The findings show that EcoKonek PH is an effective and highly usable platform for community e-waste management in Mexico, Pampanga. Users found it easy to learn, intuitive to navigate, and accessible on their existing devices, while key features such as barangay mapping, verified drop-off points, donation and redistribution, and the AI Chatbot were rated very helpful for proper e-waste disposal and digital inclusion. The strong willingness of users to continue using and recommending the system indicates that EcoKonek PH is ready for real-world deployment and can meaningfully support local sustainability and digital equity efforts.

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